



BITS Pilani Digital Design

BITS Pilani
Pilani Campus

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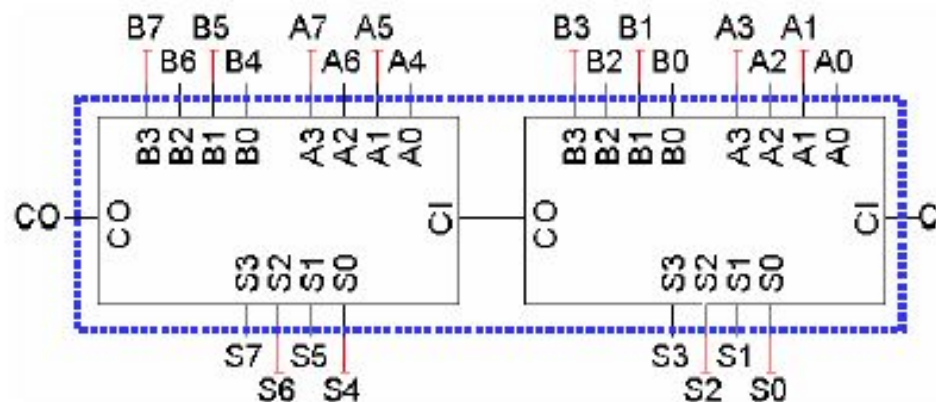
Review questions

1. Solution provided by k-maps is ...
2. Equations for Sum and Carry of a full adder

Extending the 4-bit adder: ripple carry adders

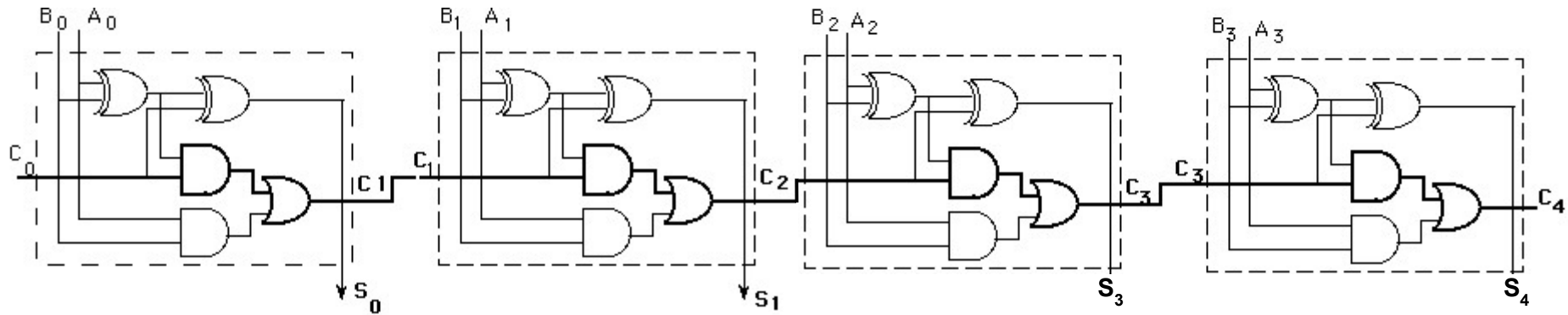


- We can use CI to combine four-bit adders together to make even larger adders, just like we combined half adders and full adders earlier.
- Here is one way to build an eight-bit adder, for example.



- How long does the addition take?

Ripple carry delays



- How much time does a 4-bit adder take to perform addition?
- How many gates does the input signal have to flow through in a single FA?
- In a 4-bit adder (count along the longest path)?
- For an n -bit adder, how many gates does the signal have to flow through?
- $2n + 1$ (for a 64-bit adder, 129 gates!)
- Assuming ~ 1 ns propagation delay per gate, how many 64-bit additions can you perform per second?
- How many do modern processors carry out?

Can we have faster adders?

- Instead of waiting for the carry out from each previous stage, we can minimize the delay by computing it directly with a two-level circuit.
- First we'll define two functions.
 - The "generate" function G_i produces 1 when there *must* be a carry out from position i (i.e., when A_i and B_i are both 1).

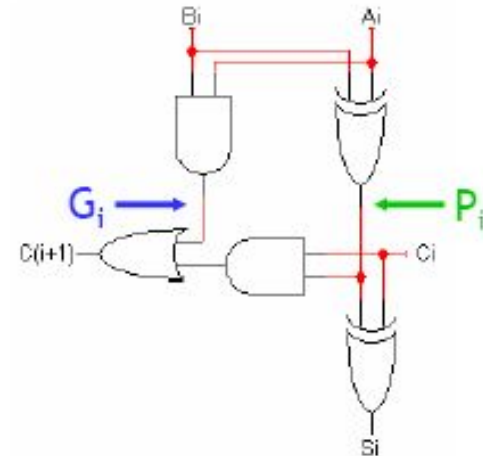
$$G_i = A_i B_i$$

- The "propagate" function P_i is true when an incoming carry is propagated (i.e, when $A_i=1$ or $B_i=1$, but not both).

$$P_i = A_i \oplus B_i$$

- Then we can rewrite the carry out function.

$$C_{i+1} = G_i + P_i C_i$$



A_i	B_i	C_i	C_{i+1}
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Can we have faster adders?

- Let's look at the carry out equations for specific bits, using the general equation from the previous page $C_{i+1} = G_i + P_i C_i$.

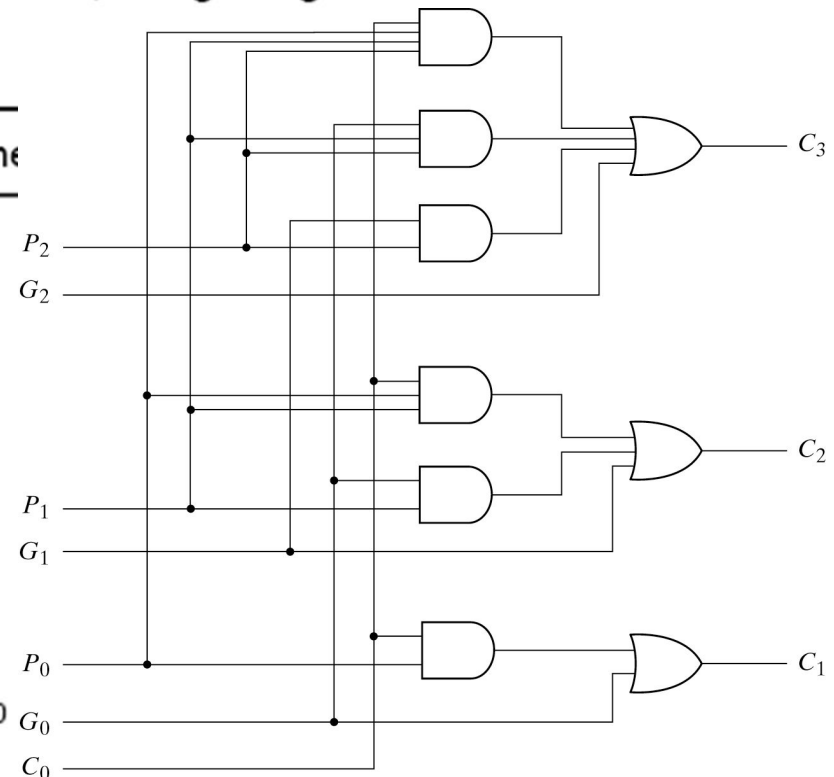
$$C_1 = G_0 + P_0 C_0$$

$$\begin{aligned} C_2 &= G_1 + P_1 C_1 \\ &= G_1 + P_1 (G_0 + P_0 C_0) \\ &= G_1 + P_1 G_0 + P_1 P_0 C_0 \end{aligned}$$

$$\begin{aligned} C_3 &= G_2 + P_2 C_2 \\ &= G_2 + P_2 (G_1 + P_1 G_0 + P_1 P_0 C_0) \\ &= G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0 \end{aligned}$$

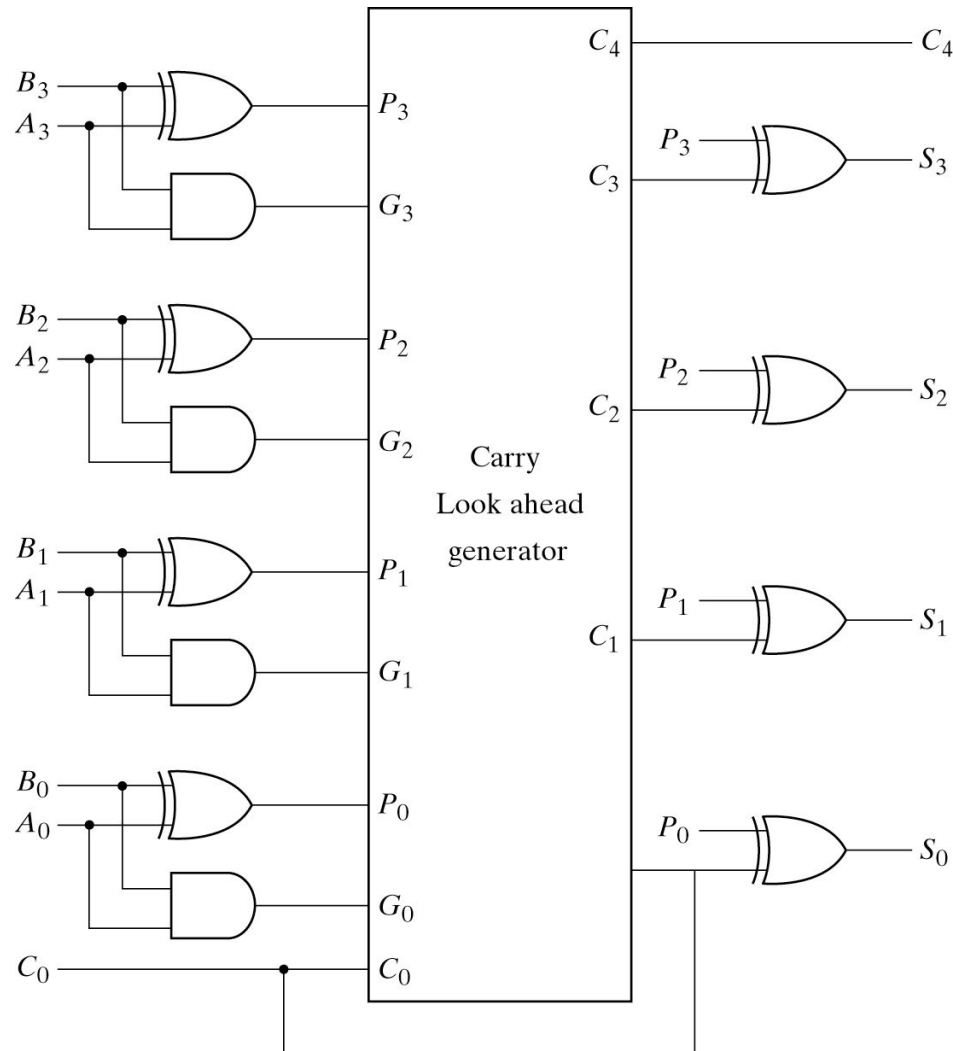
$$\begin{aligned} C_4 &= G_3 + P_3 C_3 \\ &= G_3 + P_3 (G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0) \\ &= G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0 \end{aligned}$$

Ready to see the

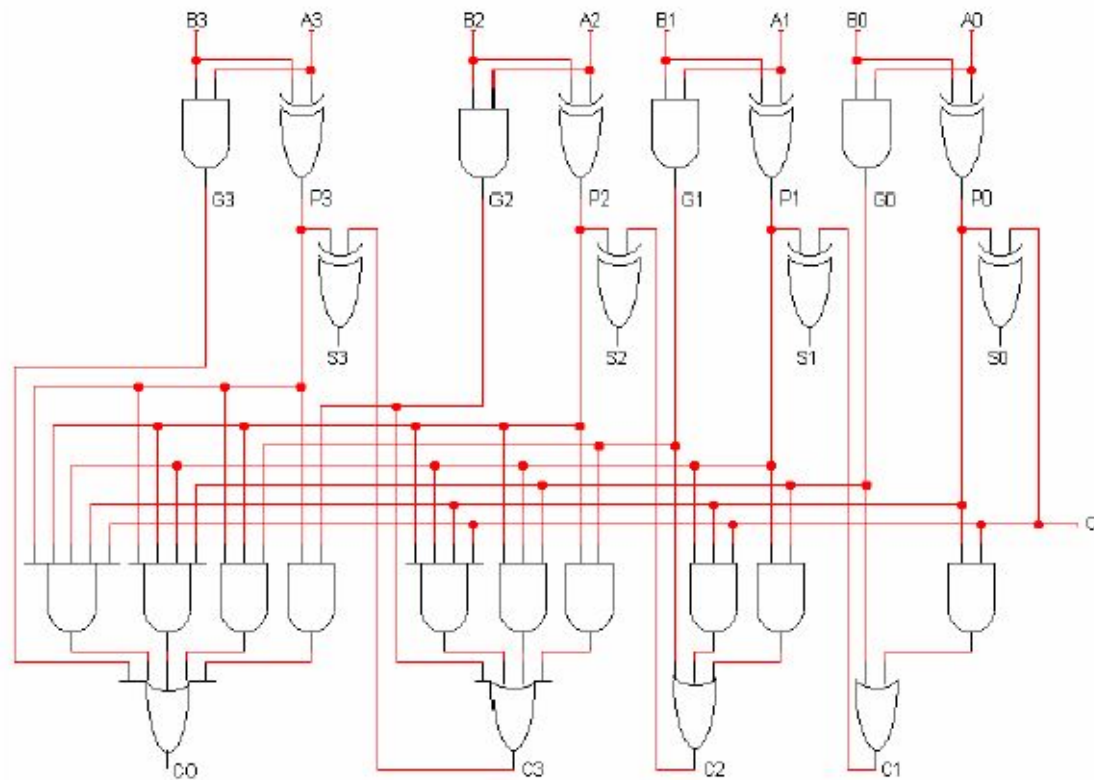


- These expressions are all sums of products, so we can use them to make a circuit with only a two-level delay.

4-bit adder with carry lookahead



A faster 4-bit adder



How many 64-bit additions can this circuit perform per second?

Building an exclusive subtractor



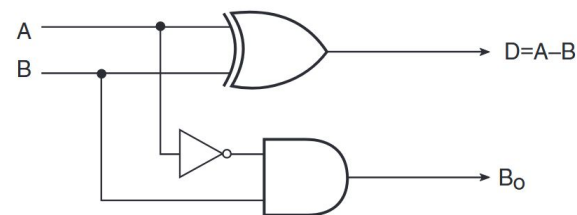
- Half subtractor

- sum $\Rightarrow$ difference
- carry $\Rightarrow$ borrow

A	B	D	B ₀
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$D = \bar{A}.B + A.\bar{B}$$

$$B_0 = \bar{A}.B$$

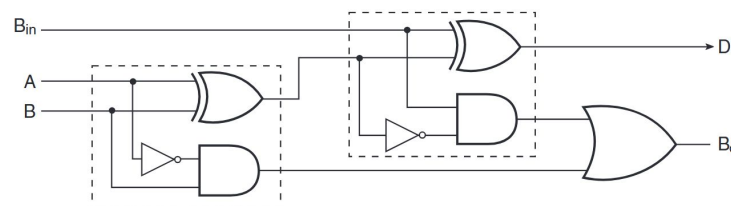


- Full subtractor

Minuend (A)	Subtrahend (B)	Borrow In (B _{in})	Difference (D)	Borrow Out (B ₀)
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$D = \bar{A}.\bar{B}.B_{in} + \bar{A}.B.\bar{B}_{in} + A.\bar{B}.\bar{B}_{in} + A.B.B_{in}$$

$$B_0 = \bar{A}.B + \bar{A}.B_{in} + B.B_{in}$$



AB \ B _{in}	$\bar{B}_{in}$	B _{in}
$\bar{A}\bar{B}$		1
$\bar{A}B$	1	
AB		1
$A\bar{B}$	1	

AB \ B _{in}	$\bar{B}_{in}$	B _{in}
$\bar{A}\bar{B}$		1
$\bar{A}B$	1	1
AB		1
$A\bar{B}$		

